INTRODUCTION

Multinomial regression is one of the most fundamental statistical models for analyzing categorical response variables with more than two classes. It plays an important role in many modern applications, including biological classification, image recognition, text categorization, and industrial quality control. In such settings, observations are typically associated with high-dimensional feature vectors, which makes classical multinomial regression difficult to apply due to overparameterization and instability in parameter estimation.

A well-known strategy for simplifying multinomial regression is the Poisson surrogate representation, often referred to as the Poisson trick, which reformulates multinomial models as a collection of Poisson regression problems. This representation establishes a close connection between multinomial regression and Poisson generalized linear models (GLMs), allowing efficient estimation through likelihood-based procedures. Recent work has further demonstrated that this representation can provide an exact or approximate likelihood equivalence under certain conditions.

However, most existing studies focus on low-dimensional settings or emphasize computational convenience rather than statistical structure. In modern applications, datasets often involve a large number of predictors, and the number of parameters in multinomial models can grow rapidly with both the number of predictors and the number of response categories. As a result, regularization techniques become essential for obtaining stable and interpretable models.

The objective of this research is to develop a high-dimensional extension of the Poisson surrogate framework for multinomial regression, grounded in the theory of generalized linear models and structured regularization. By integrating the Poisson surrogate representation with structured penalties such as group lasso, the proposed framework aims to provide a statistically coherent and computationally efficient approach for high-dimensional multinomial classification.

This work is motivated by practical classification problems involving structured categorical outcomes, such as identifying crop varieties from seed characteristics. At the same time, the methodological framework developed here is intended to be general and applicable to a wide range of high-dimensional multinomial modeling problems.

DISTRIBUTED MULTINOMIAL REGRESSION

This artical is driven by the high-dimensional-response multinomial models with large number of random counts. The original multinomial model is computationally expensive to estimate for a large number of response categories.

For document i, the joint Possion likelihood can be decomposite as

$$p(c_i) = \prod_j P(c_{ij}; e\eta_{ij}) = MN(c_i; q_i, m_i)P(m_i; \wedge_i).$$

The maximum likelihood estimate of q_i is unchanged under a variaty of sampling models for 3-way tables under the constraint that $\wedge_i = m_i$.

If we fix $\hat{\mu}_i = \log(m_i)$, which may not be the maximum likelihood estimator, then the conditional Poisson likelihood can be dealed with seperately in categories. Such strategies are Big Data approximations to their multinomial precursors.

THE POISSON TRICK FOR MATCHED TABLES

The Poisson trick is used in the context of categorical data analysis to fit binomial (multinomial) regression models by assuming independent Poisson distributions for the counts. The situation of matched tables is exemplified by the data on suicide in Germany with two classifications A and B on the gender level R. The contingency table is denoted by y^{ABR}. For two categories $R = 1$ (male) and $R = 2$ (female), we have two matched tables and total count $y^{AB}_{ab} = y^{ABR}_{ab1} + y^{ABR}_{ab2}$, which construct a binomial setting, where $\pi_{ab} = P(R = 2|A = a, B = b)$,

$$Y^{ABR}_{ab2}|Y^{AB}_{ab} \sim \text{Bio}(Y^{AB}_{ab}, \pi_{ab}).$$

"Poisson Trick": Let independent Poisson distribution $Y_{ab1} \sim P(\mu_{ab1})$ and $Y_{ab2} \sim P(\mu_{ab2})$, then joint likelihood $L(\mu) = \prod_{a,b} \frac{e^{-\mu_{ab1}} \mu_{ab1}^{Y_{ab1}}}{Y_{ab1}!} \frac{e^{-\mu_{ab2}} \mu_{ab2}^{Y_{ab2}}}{Y_{ab2}!}$ with restriction $\pi_{ab} = \frac{\mu_{ab2}}{\mu_{ab1} + \mu_{ab2}}$.

Note: The total count of the binomial model Y_{ab} is fixed or known and the count of (independent) Poisson model Y_{ab1} is random variable, as well the total count $Y_{ab1} + Y_{ab2} \sim P(\mu_{ab1} + \mu_{ab2})$.